IN THIS ISSUE

News and Developments from Arkansas, California, Georgia, Illinois, Massachusetts, Maine, Minnesota, Mississippi, New Mexico, New Hampshire, New York, Ohio, Hawaii, Rhode Island, Texas, Utah, Washington, D.C., China, England, France, India, Japan, Netherlands, Russia, Sri Lanka, United Kingdom

Volume 5 Number 2 **SPRING 1997** **© ISSN # 1072-2874**

COLD FUSION TIMES

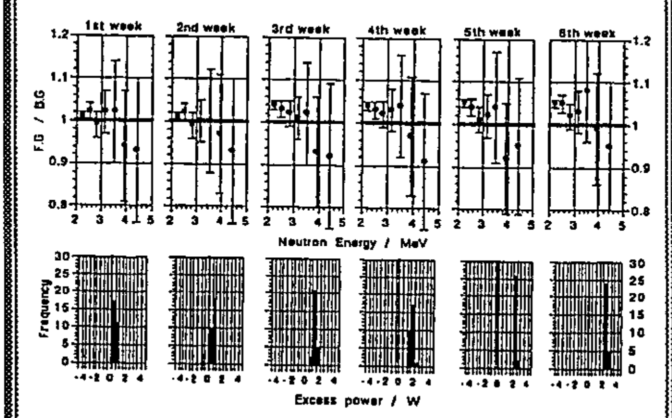
PATENT WARS: CETI's Patterson Files To Quash Drs. Pons & Fleischmann's European Patent

Sarasota, Florida - In a startling, recent development, inventor Dr. James Patterson of Clean Energy Technologies, Inc. (CETI) filed a several page complaint to block the issue of the Pons and Fleischmann cold fusion European patent (PFEP) which was about to be granted in February '97. While shock and anger rippled through the science/fusion community over this setback for cold fusion, the fact that it was created by "one of their own", was especially disheartening. Many researchers had eagerly anticipated the European issue believing it to be an important opening of the entire United States CF field, so sorely languishing in recent years. According to one insider, representatives of CETI reportedly attempted to polarize some in the cold fusion community in Sapporo, Japan (at ICCF-6) last October by asking others to join them in challenging the European FP patent application. Reportedly, there were few, if any, takers.

CETI filed their challenge just as the deadline approached, on the last week of the last month of the nine month period before PFEP would have issued, thereby opening the logjam to American cold fusion input and inventions. The biggest question on everyone's mind was "why?" and "why now?" Mr.Fred Jaeger, President of ENECO, Salt Lake City the company which bought the license to the European patent in question, sounded incredulous as he said that the CETI group's last minute challenge to the PFEP had given a "slap to the whole community."

Mr. Hal Fox, founder of the precursor to ENECO and now publisher of the *Journal of New Energy*, said that this action by CETI was "ridiculous" and "senseless internecine fighting" such that "those who have been attacking cold fusion will rejoice at this schism in the ranks of cold fusion adherents". Dr. Eugene Mallove, publisher of *Infinite Energy*, said he was saddened that "two excellent companies are seen to be fighting over turf instead of negotiating business relationships that could help everyone in the field." Another researcher remarked this appeared to be "cannibalism". Still others were struck with the paradox of how CETI initially appeared to ask to "work together" with some members Cont. page 5

Time Correlation between the Excess Neutron Emission and Excess Power Generation
[Neutrons on top (foreground/background) and XS heat on bottom]
(After H. Ogawa, Y. Oya, T. Ono, M. Aida and M. Okamoto; page 3)



Reports Of Correlated Nuclear Signatures And Excess Heat

Wellesley, MA. - This issue of the *Cold Fusion Times* includes several new reports of nuclear signatures correlated with the excess heat, supplementing the previous reports from the U.S. Navy (*Cold Fusion Times* v.5, n. 1). Dr. H. Ogawa, from the Research Laboratory for Nuclear Reactors at the Tokyo Institute of Technology, and Drs. P. Cignini and D. Gozzi, at the Universitia di Roma, report that these phenomena originate at nuclear levels. These reactions can not be the atomic level based upon ionizing radiation in the range of 89 to ~120 keV.

Cont. page 8

Electric Industry Optimistic About Dr.Frederico Pena To Head DoE

WASHINGTON D.C. - Recently the Senate voted 99-1 to confirm Mr. Federico Pena, former Secretary of Transportation, as the Secretary of Energy. Several energy groups are optimistic about the future of the DoE under Energy Secretary Federico Pena because of his reportedly favorable view toward competition within the electric power industry. Cont. page 8

TOWARD THE FUTURE

A NEW COLD FUSION MOVIE
"The SAINT" with Van Kilmer, Elisabeth Shue

The very week that "X"-files' Fox Mulder cooly discusses cold fusion as being the reason for the Majestic-12 coverup of UFOlogy, Hollywood has launched Simon Templar (Van Kilmer) as the "The Saint" who is hardly a saint until he seeks to protect the woman who has perfected cold fusion. He does this with quick change artistry, an incredible undertaking sometimes laughable, making this a well-worth "see".

COLD FUSION, RUSSIAN STYLE

If it were not for the fact that CF gets a front row seat with a newly buffed image, this film would be just another high tech action chase film with a predictable ending. But here CF, science, politics, the cold of Winter, weld a mini-education on the impact of CF to those who would otherwise stay in the dark. Simon is an international thief, millionaire, and well mustachioed master of disguise. *The Saint* is hired by the corrupt Russian dictator to steal the cold fusion formula from American nuclear physicist, Dr. Emma Russell (Elisabeth Shue). But *The Saint* falls in love with his target, who hides the secret formulae for cold fusion in her bosom over dinner, and whose life becomes in terrible danger over her discovery of the keys to making cold fusion successful.

A NEW SIMON TEMPLAR

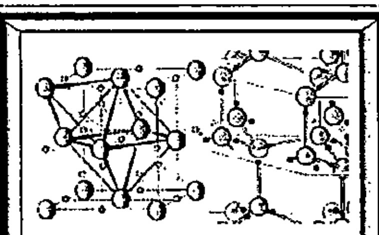
Compared to the Roger Moore television series this *Saint* has a darker Simon Templar. According to Director Phillip Noyce, "we decided Templar's story should begin in an orphanage in Hong Kong where, as a child, he learns all the tricks that he is going to use when he is an adult and how he uses these to survive. He's a man who, like a child, retreats from the brutal reality around him by using disguises which are really an escape, a mask hiding the real man."

In a key sequence of the film, Emma Russell escapes from Tretiak's agents by taking refuge from the evil son of the Russian coup lead in the American Embassy. However, the actual American Embassy in Moscow was too cramped side and situated among nondescript buildings. No problem, because they took the Peking Hotel, covered up

Cont. page 9

<<<<< Photo Credits >>>>>

Page 4 - Val Kilmer and Elizabeth Shue (*Paramount Pictures*)
Page 5 - Microspheres -- Scanning Electron Microscope image of EarthTech's duplication attempt of CETI beads (Scott Little and John Logajan); Mr. Patterson from Infinite Energy
Page 8 - Mr. Pena (Department of Energy)
Page 9 - Mr. Rothwell (Cold Fusion Times Files)
Back Cover: C/1995 O1 (Hale-Bopp) (3/10/97; Mojave desert by Bob Yen @ http://www.comet-track.com) This is a long telephoto shot, using a 6x7 camera on an AstroPhysics 610 mm f6 lens w/field-flattener. The ion tail is very strong, and extends out of the frame at least 6 deg. The dust tail has become more intense (brighter and narrower), but still has an outer fan.



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The Journal Of The Scientific Aspects Of Loading Isotopic Fuels Into Materials

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Correlation Of Excess Heat And Neutron Emission In Pd-LiOD Electrolysis

H. Ogawa, Y. Oya, T. Ono, M. Aida and M. Okamoto
Research Laboratory for Nuclear Reactors, Tokyo Institute of Technology

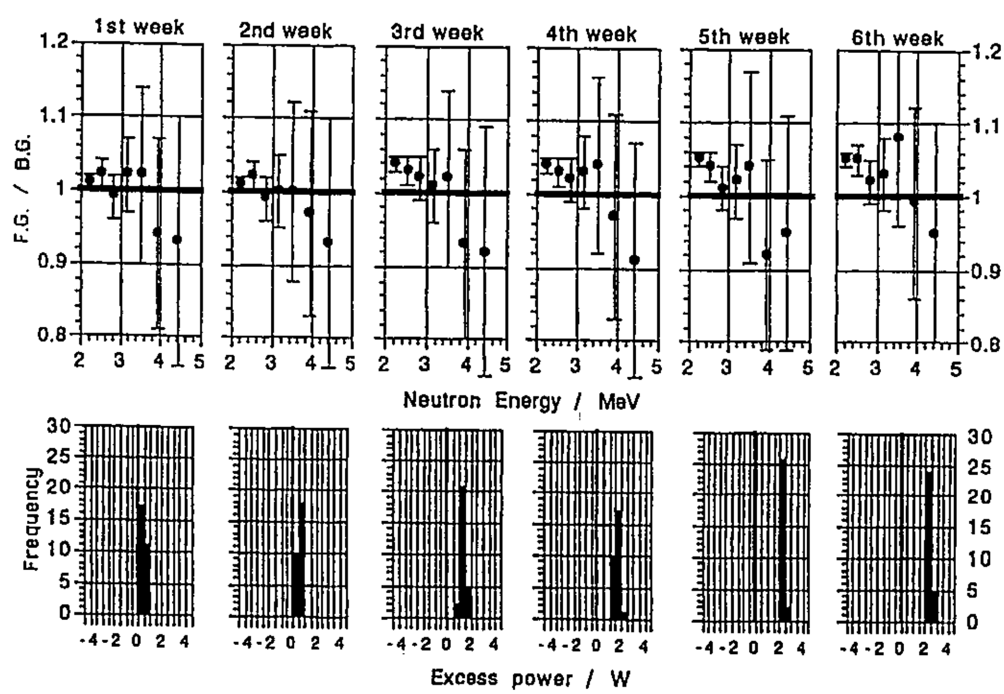
Meguro City, Tokyo Japan -- To avoid the uncertainty, especially in the excess neutron evaluation, caused from the date difference between the foreground runs and the background runs, two sets of the experimental systems have been assembled with the excess heat monitoring and the neutron detection systems of NE213. The principles of the excess heat monitoring and neutron detection are same as previously reported. The two experimental systems have been examined to determine the machine factors for the excess heat monitoring and the neutron detection in Pt 1.0M LiOH electrolysis for about one month. The machine factor for the excess heat monitoring was determined to be 1.00 and the factor for the neutron detection was determined for each ten blocks in the 1024 channels of the MCA. The neutron energy spectrum was obtained after the correction by machine factors.

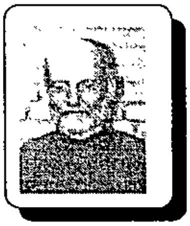
Using the two experimental systems, the electrolysis operations to observe the excess heat and the excess neutrons have been carried out coincidentally under the same conditions which gave positive results with high reproducibility as reported previously. The 3 hours/ 3 hours pulse mode electrolysis with 800 mA/20ma current density was performed with Pd - 1.0M LiOH (as the background run) and Pd 1.0 M LiOD (as the foreground run), for 5 weeks. The electrolysis cell system was also improved in the control of electrolyte level and equipped a standard heat source for the temperature calibration for the excess heat monitoring. The Pd electrodes were pretreated by fine mechanical polish and annealing in vacuo at 850 C. for 10 hours.

The excess heat up to 6.5 W (2.5% of the input power) was observed in the Pd-LiOD electrolysis and the excess neutron emission was also observed with main peak at around 4 MeV. The frequency histograms of the excess heat generation and illustrated in the figures to demonstrate the appreciable difference between the foreground run and the background run.

Time Correlation between the Excess Neutron Emission and Excess Power Generation

[Neutrons on top (foreground/background) and excess heat on bottom]





Journal Articles Which You May Have Missed

by Dieter Britz

Effect Of Ultrasound On Electrochemical Loading Of Hydrogen In Palladium

Wark A, Crouch-Baker S, McKubre M, Tanzella FL
J. Electroanal. Chem. 418 (1996) 199

The present authors decided to look at the effect on the electrolysis of water at a Pd cathode and the hydrogen loading in the metal. As reported in the old papers, the irradiation lowered cell voltage and the loading; and stopping the irradiation only restored the pre-irradiation condition after either long electrolysis, or a short burst of anodic polarization. The authors were able to tie the ca. 30-50 mV of cell voltage change upon irradiation, to adsorbed hydrogen, known to play an important role in the process at the Pd cathode.

The Phonon Mechanism Of The Cold Fusion

Liu F-S; *Mod. Phys. Lett. B* 10 (1996) 1129

Using phonon theory, the Wannier function and more, the author considers the movement of deuterons in metal deuteride as affected by acoustics. The conclusion is that predicted d-d fusion rates come to roughly observed values, near the surface where there is strong nonequilibrium and a higher electron density.

Anomalous Thermal Power Generation From Proton-Conducting Oxide

Oriani RA; *Fusion Technol.* 30 (1996) 281

This is a high-temperature (ca. 400C) calorimetry experiment, using a solid state electrolyte, perovskite $SrCe_{0.9}Y_{0.08}Nb_{0.02}O_{2.97}$, an ion conductor, supplied by Mizuno. A Seebeck-effect calorimeter of refined design was used for accuracy. The solid electrolyte was simply heated at a known power, and deuterium or helium (as a control) allowed into the chamber, monitoring the heat given off. There appears to be clear evidence of up to 4-σ excess power (relative to noise) with deuterium, but never with helium. Small dc power currents were applied to the electrolyte disks, but the results show no clear effect correlating with this. The success rate was low, and so were the excess powers.

Recent Results And Activities On The New Hydrogen Energy ("Cold Fusion")

Takahashi A; *Suiso Enerugi Shisutemu* 21 (1996) 39

The author believes that CNF has been demonstrated, excess heat found but without fusion taking place; and that some unconfirmed reports claim helium^4 and should be repeated. The key is to pin down the nuclear or chemical origin of excess heat.

New Approach Toward Nuclear Fusion Without Strong Nuclear Radiation

Li X *Nucl. Fusion Plasma Phys.* 16 (2) (1996) 1

Li goes through some QM theory and concludes that lattice-confined ions react in a different way from beams hitting a target. Because of the Coulomb barrier, only the long-lived energy levels have a chance to resonate, and thus (fast) reactions emitting neutrons do not occur; instead, only those not emitting neutrons do occur, which supports the cold fusion claim.

(in Chinese).

Anomalous Isotopic Distribution Of Elements Deposited On Palladium Induced By Electrolysis

Mizuno T, Ohmori T, Kurokawa K, Akimoto T, Enyo M; et al.
Denki Kagaku oyubi Kogyo Butsuri Kagaku 64 (1996) 1160

This was a long-term (one month), high-current-density electrolysis experiment in a heavy water electrolyte (LiOH and Li_2CO_3), and surface analysis using several methods, before and after electrolysis. The usual forest of peaks is found, as expected from long electrolysis (and previously found by others); the authors checked for isotope ratios, however, and found some that deviated significantly from the normal values and concluded that this shows that some isotopes were produced during electrolysis. They go on to speculate on possible nuclear mechanisms to fit the data.

(in Japanese)

Anomalous γ–Peak Evolution from SrCe Solid State Electrolyte Charged in D_2 Gas

Mizuno T, Inoda K, Akimoto T, Azumi K, Ohmori T, et al.;
Int. J. Hydrogen Energy 22 (1997) 23

This team "electrolyzed" disks of the high temperature ionic conductor consisting of mixed oxides of Sr, Ce, Y and Nb, sintered into disks at 1300-1480 C and coated with Pt on both sides for contact. Voltages up to 5-45 V and frequencies of 0.0001-1 Hz were then applied to these disks ("electrolysis") in atmospheres of deuterium or hydrogen, and a gamma detector was used to measure the gamma spectrum emitted. These showed peaks that were not there before electrolysis, and the authors assign them to various isotopes such as ^{197}Pt, ^{153}Sm and ^{155}Sm, thought to have been produced during the experiment. They go on to surmise various nuclear reactions that might lead to these isotopes and speculate that they may be the key to understanding cold fusion.

visit Dieter Britz's site at http://kemi.aau.dk/~britz/

Fukuhara M (Toshiba Tungaloy Co Ltd); *Jpn. Kokai Koho*
JP 08,194,079, 13-Jan-95 Cited in *Chem. Abstracts* 125:259415 (1996).

** "A D containing. alloy (made of >= 1 of metal of Group IV or VIII elements such as Pd and/or Ti) at a temp. lower than room temp. (between room temp. and liq. N temp.) is irradiated with dynamic ultrasonic wave, thereby the alloy is brought to a state of nonequil. high-temp. and high-pressure to induce a reaction, the resulting excess heat is transported to a heat exchanger by a pressurized medium (such as pure H_2O or He gas) and is circulated for using. A large amt. of heat can be generated easily and economically without air pollution and waste production.

PEOPLE IN THE NEWS

Dr. Patterson

CETI Files To Quash P&F Patent

continued from Page 1

of the CF community, only to then make this move after reportedly seeking to recruit allies in their imminent PFEP challenge at Sapporo.

CETI'S VIEW

Dr. Patterson (Ph.D. in Science, Pacific Western University, LA), inventor of some of the first American cold fusion patents to issue, was unavailable for comment but the business manager and President of CETI clarified their view for the *COLD FUSION TIMES*. According to CETI business manager Mr. Christian Ismert [Sarasota office], Dr. Patterson and CETI officials felt this move would be a "boost for the field". Mr. Ismert admitted that the challenge and blockage of the PFEP would actually "not be considered a friendly move by some" but that CETI was "doing what is in the best interests of our investors."

CETI President Jim Reding stated that CETI had "issued the challenge" to prevent "anticompetitive reflections" which could have hurt the field. He stated that CETI felt the FP claims were too broad and would thus inhibit development in the field; but by challenging the FPEP there would be "benefit to the field" because the FPEP claims were "very broad without teaching adequately" how to use cold fusion. "The laywers saw it right away", Mr. Reding said.

Mr. Reding also noted that the CF issue involving the nickel-light water/palladium-heavy water dichotomy -- often ignored by the hot fusion critics -- was actually also an issue. According to Mr. Reding, Dr. Fleischmann and Pons had stated in 1989 that light water excess heat does not exist, but Mr. Reding noted that even so, the FPEP claims attempted to also claim nickel reactions. Thus, given that Europe is "the litmus test for the rest of the world" at this point in time, and given that it would considerable more expensive to begin the legal fight later, and because they did not want to "pay for a technology that did not benefit anyone" (the ENECO-held FPEP claims), Mr. Reding said that Dr. Patterson had to act now.

Despite the grim response in the scientific community, this current CETI tactic versus the PFEP must have created one large, wry smile on the faces of some legal counsel and marketing executives. Some scientists contacted for this article, have reasoned that this surreptitious move -- to cripple the European patent issue for Dr. Fleischmann and Pons -- was forged on the dream of a financial empire because one problem preventing additional funding for CETI may have been the European Fleischmann Pons application.

An alternate hypothesis floating in CF circles is that Dr. Patterson now wants "credit" for work previously acknowledged to belong to Drs. Fleischmann and Pons. Worldwide, Flesichmann and Pons, are not only the acknowledged pioneers of this field, but the scapegoats as they bore the brunt of the slings and arrows thoughtlessly thrown at them. According to Mr. Reding, CETI apparently recognizes the FP discovery and March 1989 announcement of CF, but Mr. Reding insists that the FPEP challenge is only on the claims issue and not the original discovery of the effect.

Whatever their claimed motivations, CETI has also begun other legal suits. According to one experimenter into this field since 1989, his associates have been on the receiving end of a "letter battle" with CETI's lawyers merely for posting an ad in *Infinite Energy* offering to sputter-coat polystyrene beads with metals for use in cold fusion systems. He states that sputter coating plastics is a long established technique, and has found the purported hasselling very disconcerting. CETI's Reding states "they are violating our patents".

DR. PATTERSON'S PATENTS ISSUED

Dr. Patterson filed his patents (now assigned to CETI) after the seminal cold fusion announcement of March 1989. It has been noted by those familiar with patents in the CF field that Dr. Patterson's patent counsel rather skillfully maneuvered the application in an extraordinary manner during the patent process. The Patterson patent application, according to some involved with this matter, circumvented the US Patent Office's normal patent review process involving the standard nuclear physics group by claiming Dr. Patterson's vulnerability vis a vis his age and health.

Thus, his application avoided the dreaded "pro-hot fusion" and "anti-cold fusion" Group Art 220, and his application reportedly drifted within the Patent Office to what is considered the more relatively benign non-nuclear patent examiners in Group Art 110. In the words of one individual familiar with the process, Dr. Patterson's patent was issued before Group Art 220 even "knew what was going on".

By such means of apparent reported relative reliance upon Dr. Patterson's age and/or health rather than usual scientific rigor, Dr. Patterson's patent application by-passed the tough and lengthy procedure all others have had to transverse, and he was ultimately issued a US patent. Soon after Dr. Patterson was issued a US patent on his covered beads, there followed reports of "kilowatts" of power with a few watts input in the media market including ABC-TV news (*Cold Fusion Times* v. 4, n. 2).

Cont. page 8

CETI Multilayer Thin-Film Microspheres

(Right top) The multilayers of the Patterson microsphere.
(Left) The CETI setup of vertical flow calorimetry used by Cravens, Patterson, and CETI
(Below) The heterogeneity of a collection of a group of microspheres pedigree unknown.

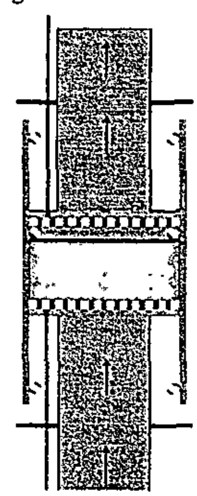
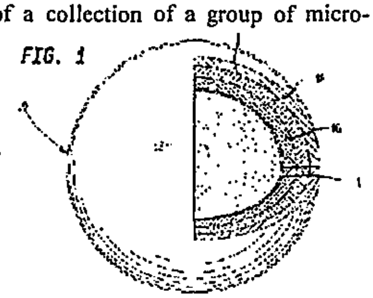
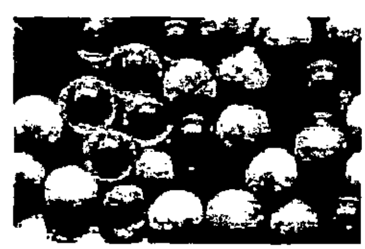


FIG. 1





COMING:

More Technical Papers
Nickel, Titanium, Palladium
Additives
Isotopic Shifts
Survey Results
Gas/Glow Discharge Loading
Lavoisier & Platinum Metallurgy
Nanodot Technology

Lavoisier melting a "worthless" platinum nugget

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The **COLD FUSION TIMES** wants you. Your subscription and support is needed. It is the only cold fusion journal which continues to be devoted exclusively to the material, economic, and intellectual property issues of loading isotopic fuel - a hydrogen isotope - into a material used to produce useful energy. Thank you for your past support.

X-RAY, HEAT EXCESS AND He^4 IN THE ELECTROCHEMICAL CONFINEMENT OF DEUTERIUM IN PALLADIUM

F. Cellucci, P.L. Cignini, G. Gigli, D. Gozzi, M. Tomellini,
E. Cisbani, S. Frullani, F. Garibaldi, M. Jodice, G.M. Urciuoli
Dipartimento di Chimica, Universitia di Roma
Laboratorio di Fisica, Istituto Superiore di Sanita
Continued from page 10

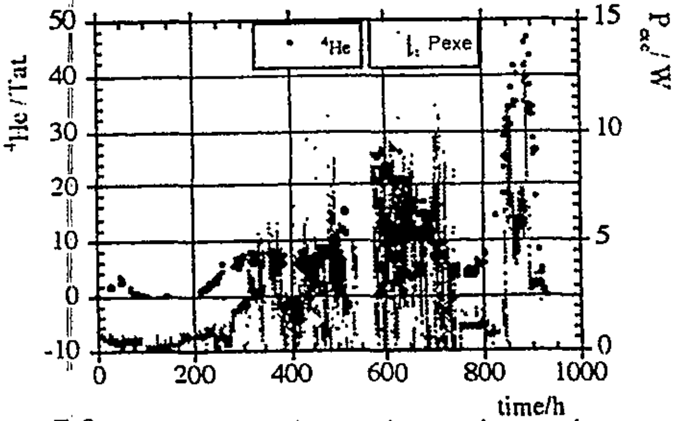
also continuously monitored the Ne^{20} signal. At set intervals (equal or greater than 900 s), a sample of 150 cm^3 of the electrolysis gas mixture (N_2, D_2, O_2, D_2O vapor, He^4 if any), escaping from the calorimeter-electolysis cell is made to flow through a deuterium cutting system. It is sucked through a LN_2 charcoal trap which absorbs almost all the gases except He^4. This is injected at low pressure ('less than equal to' 5 x 10^{-2} mbar) into the QMS chamber where the pressure is generally 'less than equal to' 1 x 10^{-7} mbar.

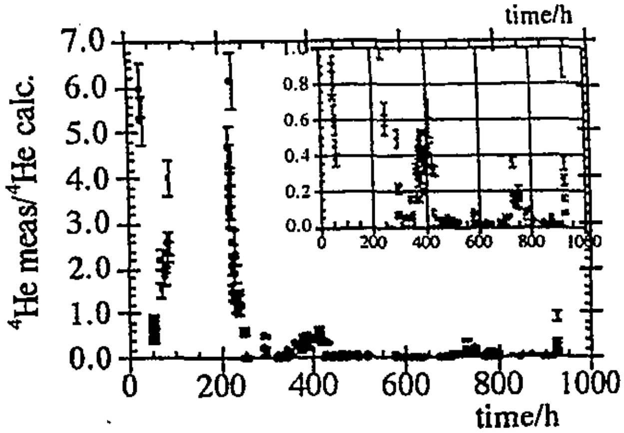
In a four-cell (5) experiment lasting 950 hours, more than a thousand (1,000+) samplings were performed. For each cell, a satisfactory time-resolved structure of He^4 release data was obtained. Then all this data has been correlated by a simulation procedure to the measured excess heat by computing the expected He^4 at the time when the sampling was effectively done.

Further, a very important result ... is a clear-cut evidence this phenomena we are dealing with is of no chemical nature. An x-ray film positioned 5 cm from a cell has been impressioned with a series of spots roughly reproducing the image of the cathode. Another x-ray film located in the same position at the same time and for the same period in front of a blank cell was not impressioned at all. Through microdensitometry of the exposed film and calibration of the optical density against the x-ray exposure, each spot characteristics was recorded by position, dimension and intensity.

Since all the geometric parameters and nature of all materials interposed between the cathode and film are known, it has been possible to calculate the exposure as a function of the energy of radiation and the I/Io ratio. The spot nature of the impressioned film is very intriguing and it could support more than one explanation about the main process occurring in the source. But in the absence of a clear understanding of it, in order to estimate the value of the involved energy, we are oriented (inclined?) to treat the matter in terms of classical physics of radiation. This is by taking account the cathode

Calculated Helium-4 (left side) and excess heat power (right side)





Ratio between measured and calculated Helium-4 versus time. The insert (located in the upper right) is a magnified view in the range from 0 to 1.

structure. The structure was a bundle type having 150 Pd wires of 250 μm diameter.

The analysis of the data leads to the conclusion that (because the energy of radiation and the total energy associated to all the spots on the whole solid angle were, respectively, evaluated to be (89 $^{+/-1}$) keV and (120 $^{+/-0.4}$) kJ), **we are dealing with a phenomena which originates at the nuclear level and not at atomic level** (soft gamma rays instead of hard x-rays) and it is not the principal source of energy (being approx. 0.5% of the energy measured by calorimetry in the same interval of time.

next issue: ***COLD FUSION TIME'S***
TECHNOLOGIES
COLUMN RETURNS WITH NEW MATERIAL

Group VIII
Atom Clusters
More on Helium Production and
Other Nuclear Signatures
Secondary to Cold Fusion

COLD FUSION TIMES

ENGINEERING AND MATERIALS VIEW TOWARD COLD FUSION

Design Considerations For Multilayer Thin-Film Patterson-Type Microspheres

G.H. Miley, G. Narne, M.J. Williams, J.A. Patterson,
J. Nix, D.Cravens, and H. Hora
Fusion Studies Laboratory, University of Illinois

Urbana, Il. The selection of thin-film metal parts is based on four parameters: Maximum Fermi energy level difference, good hydrogen/deuterium (H/D) diffusivity, good H/D 800 Angstroms thick films of Pd/Ni and Pd/Ti, giving Fermi level differences (ΔF) of 1.5 and 1.7 respectively. These combinations were selected on the basis of ready availability of materials and ease of manufacturing, but other combinations potentially offer more intense reaction rates. Examples include Ce/Be, Pd/Zr, Th/Pt, and Th/Be, with ΔF = 7.4, 1.6, 2.4, and 5.5 ev, respectively.

Experimental Observation Of Massive Transmutation In Multilayer Thin-Film Microspheres

G.Miley, G. Narne, M.Williams, J.Patterson, J. Nix, D. Cravens
H. Hora Fusion Studies Lab, U of Illinois

Urbana IL -- Over a dozen experiments have been carried out using multilayer thin-film microspheres, designed on the basis of "swimming electron layer" (SEL) theory[1,2] using a 1 molar lithium sulfate light water electrolyte in a flowing packed-bed electrolytic Patterson (TM) - type cell. Significant quantities of transmutation products have been obtained in all of these runs, which have lasted periods of time ranging from one to five weeks. Transmutation of the metallic coatings to new reaction products, with yields exceeding 50% of the initial metal weight, have been obtained in some cases. Excess heat is also observed in these runs, but varies greatly, depending on the configuration. The largest yields have been obtained for Cu, Al, and Ag, although a variety of additional products are observed, using a combination of secondary ion mass spectrometry (SIMS) analysis and neutron activation analysis (NAA). The dominant product varies depending on the specific coatings employed, with the largest yields typically coming from single-layer Pd or Ni on cross-linked polymer microspheres or from multilayer Ni-Pd-Ni-Pd-Ni microspheres.

While the yields are so large as to preclude the possibility of impurity artifacts, some special runs have been carried out and a unique "clean" cell, where the only metal present, in addition to the microspheres, were the Ti, or Pt electrodes. Both heavy and light products, with atomic numbers considerably above and below that of the base metal (Ni or Pd), are observed.

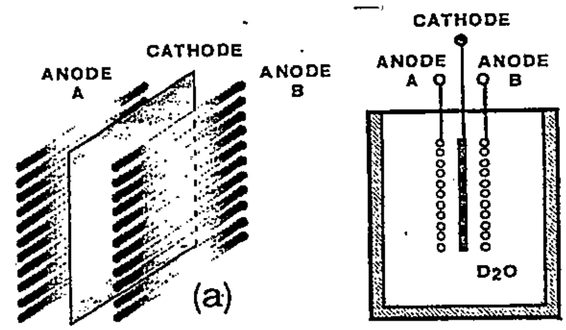
This can only be explained by proton-induced transmutations, followed by a fission of the heavier-mass product. The resulting process is quite unique and has been termed "filfiss".

1. G.H. Miley, H. Hora, E.G.Batyrbekov, R.L. Zich, "Electrolytic Cell with Multilayer Thin-Film Electrodes", Fusion Tech. 26, 4T, part 2, 313-320 (1994).
2. H. Hora, J.C. Kelly, J.U. Patel, M.A. Prelas, G.H. Miley, and J.W. Thompkins, "Screening in Cold Fusion Derived from D-D Reactions", Physics Letters A. 175, 138-143 (1993).

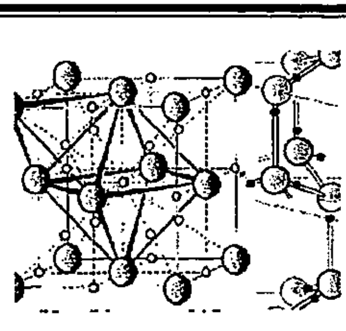
Triode Cell Experiments For Controlled Fleischmann/Pons Effect

Evan L.Ragland
The Boiler Works

Diamondhead, MS - Experimental research and evaluation of three electrode (triode) cold fusion electrolysis cells is reported herein. Apparatus development began, after patent application, in June 1995. The triode apparatus introduces controlled loading and operation of Fleischmann/Pons-type cells. In August 1995 excess heat generation was observed in triode apparatus experiments by Dennis Cravens in his laboratory in New Mexico. In November 1995 the Boiler Works laboratory in Diamondhead began experimental evaluation of the triode apparatus. Understanding gained from these experiments led to development of a triode fusion reactor. The reactor has been in continuous operation since 20 March 1996.



The experimental reactor data base is being applied in further triode apparatus developments. Near-term goals are, completion of a reactor test bed for "quick change" cathode specimen evaluation and engineering design of a 5 to 10 KW reactor cell. Thin film cathode specimens prepared by the Materials Science and Engineering Laboratory of the University of Alabama in Birmingham are presently ready for test and evaluation. These include Pd film on Ag, Al, Cu. and quartz substrates and Pt films on Si bead specimens.



The Journal Of The Scientific Aspects Of Loading Isotopic Fuels Into Materials

Reports Of Correlated Nuclear Signatures And Excess Heat

cont. from page 1

Autoradiography is used again to further demonstrate the nuclear basis of these reactions. The discovery of linked excess heat and minuscule neutron production, and also excess heat with moderate levels - but linked - helium-4 production are consistent with other cold fusion reports.

Industry Optimistic

cont. from page 1

His previous role creates a historic link between energy, environment and transportation. Mr. Pena earned his Bachelor's degree (University of Texas, 1969) and Law Degree (University of Texas, 1972), and served as Mayor of Denver, Colorado from 1983 to 1991.

Congress held up his nomination over mainly one concern. Republican concern have centered upon nuclear waste issues over how to dispose of thousands of tons of radioactive waste from nuclear power reactors. Congress and Secretary Pena would be well advised to study the possible impact of cold fusion upon some aspects of this problem, too. There is a role for remediation as well as energy production. "In addition to revamping nuclear waste programs and dealing with this nation's dangerous dependence on foreign oil, [Secretary Pena] faces calls from Congress and the public to shut down the department," said Senate Energy Committee Chairman Frank Murkowski, an Alaska Republican. "Secretary Pena faces many challenges, not the least of which is stopping the pile-up of high-level radioactive waste at 80 locations in 41 states." Regarding the interim storage problem, Sen. Murkowski has pushed a bill to build a temporary storage site in Nevada to take the spent fuel, which the government legally becomes responsible for in 1998.

CETI Files vs P&F Patent

continued from Page 1

Many scientists still await, however, the complete series of normal peer-reviewed scientific publications. Meanwhile, CETI has begun what is perceived by some as the first lobbing of nuclear weapons in the cold fusion story against at least two parties. As one savvy engineer, familiar with the Patent process and the means used by Dr. Patterson to bypass Group Art 220, said -- after being informed of the most recent CETI maneuver in challenging the PFEP -- "He (Dr. Patterson) must be feeling better now".

THE "NEW HYDOROGEN ENERGY"

The word "hydorogen" is purportedly how the Japanese decided to spell, or did misspell, hydrogen. The misspelling graces the cover of the ICCF-6 proceedings, produced by the Japanese CF scientists.

(Left) Reproduction of the Cover of the ICCF-6 Proceedings
(Right) NHE later sent out this letter like this with a package of stickers to cover up over the ICCF-6 title misspelling.

The Sixth International Conference on Cold Fusion

PROGRESS
IN NEW
HYDOROGEN
ENERGY

Edited by Makoto OKAMOTO

To All Participants:

Enclosed, please find the stiker to correct the title of the proceedings which we sent you before. Please stick it on the title of the coverpage. We are very sorry to give you a trouble.

Sincerely yours,
Secretariat ICCF-6

EVENTS IN JAPAN -- NEDO

In the last issue, we presented Jed Rothwell's letter to the Japanese NEDO corporation. Here is one of his recent Internet postings, and the rapid response from the organization itself.

Jed Rothwell

Concord, New Hampshire -- "Rumor has it that the NEDO cold fusion program has finally bit the dust. They have reportedly cut off funding to SRI, and they are preparing to shut down the operation in Japan too. Selling off equipment, giving people their walking papers.

I have not confirmed this at the highest levels yet, but it is what I have been expecting for some time. As I reported here and in *Infinite Energy*, during ICCF6, the NEDO managers told me the program was in deep trouble and they were not sure of additional funding. The fiscal year in Japan ends in April, so now is when the money runs out. (That is also the end of the academic year, by the way.)

All I can say is: I Told You So. I mean Fleischmann, Storms, Cravens, Ikegami, Bockris and many others told them so. I cited them in my Open Letter. But the NHE went on for six more months ignoring the best advice from the top experts in the field. So what else could have happened? This is not hot fusion; we do not a free ticket to screw up, year after year, producing no results. Here in the real world when you set a goal and you repeatedly fail to meet it, out you go. Academics and bureaucrats get to thinking they have a sinecure, until it is too late. Then they blame me for warning them!

Elliot Kennel has already accused me of being in league with Huizenga and Morrison. He believes in shooting the messenger rather than paying heed to the warning. I guess it is up to us now. If we cannot pull off a good demo, cold fusion will die, maybe for years, maybe forever. Many other nascent technologies have withered on the vine."

- Jed Rothwell (*Infinite Energy*)

Sapporo Japan -- "Oh, please! I only work for NEDO, whereas I've already conceded to you the role of Official Last Word in Cold Fusion, so perhaps I'm not qualified to comment on the rumors you have heard. But I always hate rumors from unnamed sources anyway. As I understand the situation, the four-year project was originally scheduled to terminate at the end of March, but has been extended for another year at a lower funding level than last year. So you can interpret that negatively or positively as you choose.

However, in my opinion this is the number one place in the world for cold fusion research. That's why I'm here (BTW, if there is a place in the US that can compare, let me know and I'll send my resume there quick). In the meantime, NEDO extended the fellowship for me and for Andrei Lipson until April 1998.

We are also continuing to sponsor research in electromigration with the Celani group at Frascati, Italy and now have an operating cell here, for which yours truly is the group leader (although Hiroshi Kamimura is the top technical specialist here in pulsed electronics, and has done most of the work). I wish you nothing but success in your technical endeavors. Besides, if your judgments have been correct to date, by simply following your own suggestions, success ought to be inevitable."

- Yours truly, Elliot Kennel

The "Saint" cont. from page 1

the Communist insignia, took away all the Chinese lettering, put up a gate and, voila, a pseudo-American embassy. Viewers will thank Gene Mallove for moving the movie story toward the actual cold fusion reality as he contributed as a consultant.

Readers of *The Saint* go back nearly seventy years, when British author Leslie Charteris first created the character of Simon Templar, The Saint, in the story "*Meet the Tiger*" (1928).

WORLD WIDE WEB SOURCES ON COLD FUSION

• **COLD FUSION TIMES**' home page:
http://world.std.com/~mica/cft.html

- Dieter Britz's home page http://kemi.aau.dk/~britz/
- John Logajan's home page: http://www.skypoint.com/members/jlogajan
- *Infinite Energy* home page:
http://www.mit.edu:8001/people/rei/CFdir/CFhome.html
http://ourworld.compuserve.com/homepages/JedRothwell/

American Nuclear Society Conference
June 1 - 5 Orlando, Florida.
Cold Fusion Session Wednesday, June 4
"Low Energy Nuclear Reactions."
Panel discussion: George Miley, Hora and Patterson.

TO COLDLY GO WHERE NO ONE HAS GONE BEFORE

COLD FUSION TIMES

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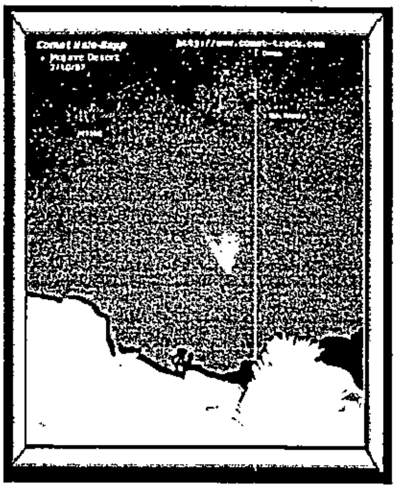

(left) C/1995 O1 (Hale-Bopp)
photo credit page 2

Correlations of excess heat and nuclear signatures highlight this issue. Here from the Departmento di Chimica in the Universitia di Roma comes further linkage of ionizing radiation, excess heat, and helium-4 production.

X-RAY, HEAT EXCESS and He^4 IN THE ELECTROCHEMICAL CONFINEMENT OF DEUTERIUM IN PALLADIUM

F. Cellucci, P.L. Cignini, G. Gigli, D. Gozzi, M. Tomellini, E. Cisbani, S. Frullani et alia
Dipartimento di Chimica, Universitia di Roma

Roma, Italy After seven years of worldwide research activity in the field of the cold fusion the energy balance of the observed anomalous phenomena is still far from to be completely proved. This contributes to maintain skepticism in the scientific community. We believe that the best arguments to support the field can arise from those experiments in which excess heat and nuclear evidences are correlated. Since 1989 our activity was addressed to this end with a continuous effort to improve the reliability of the measurements. We dedicated particular attention to neutron detection and due to the high efficiency and overall quality of the system the negative results have found the conclusion that, even in agreement with other reported findings, the neutrons, as well as the tritium production, are very low probability channels in the overall process. Furthermore, a very great effort was expended to obtain reliable on-line measurements of He^4. This was finally obtained by a 'home-made' system based on a high resolution and high sensitivity (=7pA/10^{12} atoms of He^4) quadruple mass spectrometer (QMS). It was connected to an on-line automatic sampling device where the occurrence of air contamination was definitely avoided and,

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TO: